PERIODIC TABLE AND ELECTRONIC CONFIGURATION

PYQ • 2021 - 2026

1. 2026 **3 MARKS**

(A) The last 2 electrons of the element P belonging to the third period are filled in the s subshell. (The symbol is not real)

(a) What is the atomic number of this element ?

(b) Write the chemical formula of the oxide of this element.

(c) Write the subshell electron configuration of the element in the fourth period belonging to the same group.

OR

(B) The position of two elements in the periodic table is given. (The symbols are not real)

Element P : Period – 2, Group – 16 Element Q : Period – 3, Group – 1

(a) Write the n and l values of the outermost subshell of Q containing electrons.

(b) How many orbitals will be present in the subshell containing the outermost electron of P ?

(c) Write the chemical formula of the compound formed by combining P and Q.

2. 2026 **2 MARKS**

Manganese (Mn) is an element that exhibits different oxidation states. (Atomic number of Mn – 25)

(a) Find the oxidation state of Mn in MnO_2.

(b) Write the subshell electron configuration of manganese ion in MnO_2.

3. 2025 **4 MARKS**

Manganese (Mn) is an element that belongs to d block in the periodic table.

(a) Outer subshell electronic configuration of Mn is $3d^5 4s^2$. Find the atomic number of Manganese.

(b) Find the oxidation state of Mn in Mn_2O_7.

(c) Write period number and group number of Manganese.

(d) Write the subshell electronic configuration of Mn^{4+} ion.

(Hint : Oxidation state of Oxygen = – 2)

4. 2025 **3 MARKS**

Subshell electronic configuration of an element 'A' is given in two different ways. (Symbol is not real)

(i) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^2$

(ii) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2$

(a) Which of these is the correct subshell electronic configuration ? Why ?

(b) Identify the block in the periodic table to which this element belongs ?

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5. 2025 **1 MARK**

Many of the metals in _____ block are used as catalysts in petroleum industry.

(s, p, d, f)

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6. 2024 **4 MARKS**

The subshell electronic configuration of an element is given :

$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^6\ 4s^2$

(a) What is the atomic number of this element ?

(b) Which subshell among these has the highest energy ?

(c) Find out the period number and group number of this element.

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7. 2024 **2 MARKS**

(a) Oxidation state of Mn in MnO_2 is +4. Find out the oxidation state of Mn in Mn_2O_3.

(Hint : Oxidation state of oxygen = – 2)

(b) Transition elements show variable oxidation state. Why ?

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8. 2024 **1 MARK**

f-block elements which belong to 7^{th} period are known as _____ .

(Transition elements, Lanthanoids, Halogens, Actinoids)

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9. 2022 **5 MARKS**

Atomic number of Manganese (Mn) is 25.

(a) Write the subshell electronic configuration of Mn.

(b) Find the block of Mn in the periodic table.

(c) Which category of elements does Mn belong ?

(Transition element, Halogens, Noble gases, Alkaline earth metals)

(d) What is the oxidation number of Mn in MnO_2 ? (Oxidation number of oxygen is – 2)

(e) Write the subshell electronic configuration of Mn^{2+}.

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10. 2022 **3 MARKS**

(a) Atomic number of an element is 17. Write its subshell electronic configuration.

(b) Find the group number and period number of this element in the periodic table.

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11. 2022 **1 MARK**

How many electrons are donated by first group elements generally in chemical reactions ?

(1, 2, 3, 4)

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12. 2022 **1 MARK**

Which one of the following subshells has the highest energy ?

(1s, 3d, 4s, 3p)

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13. 2021 **4 MARKS**

The subshell electronic configuration of an element is given.

$1s^2\ 2s^2\ 2p^6\ 3s^2$

(a) Write the atomic number of this element.

(b) To which block this element belongs ?

(c) Identify the period number and group number of this element.

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14. 2021 **3 MARKS**

Outermost subshell electronic configuration of two elements X and Y are given.

X = $3s^2$ Y = $3s^2 3p^5$ (Symbols are not real)

(a) What is the valency of the element X ?

(b) Which element shows metallic character ?

(c) Write the chemical formula of the compound formed by the combination of X and Y.

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15. 2021 **3 MARKS**

The chemical formulae of two different chlorides of iron (Fe) are given below.

(i) Ferrous Chloride – $FeCl_2$ (ii) Ferric Chloride – $FeCl_3$

[Hint : Oxidation state of chlorine atom (Cl) = – 1; Atomic number of iron (Fe) = 26]

(a) In which compound iron (Fe) shows +2 oxidation state ?

(b) Write the subshell electronic configuration of Fe^{3+}.

(c) Why does iron show different oxidation states ?

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16. 2021 **2 MARKS**

(a) From the given subshell electronic configuration, write the correct electronic configuration of Chromium (_{24}Cr).

(i) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^4 4s^2$

(ii) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$

(b) What is the reason for this selection ?

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17. 2021 **1 MARK**

What is the maximum number of electrons that can be accommodated in 'f' subshell ?

(2, 6, 10, 14)